

Voice Based Concept Understanding Analyser Using Artificial Intelligence

Project Report

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

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CERTIFICATE

This is to certify that the project entitled "Voice Based Concept Understanding Analyser Using Artificial Intelligence" is a bonafide work carried out by **Swetha Madda, Syed Mohammad Saad, Yogeswari Vasantala, Shiva Shankar Yellaboyina** in partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Computer Science and Engineering during the academic year 2026–2027.

The work presented in this report has been completed under the guidance of the project mentor and has not been submitted elsewhere for the award of any other degree or diploma.

Project Guide

Head of Department

Principal

Date:

Place:

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Abstract

The Voice Based Concept Understanding Analyser Using Artificial Intelligence is an intelligent web application developed to evaluate a student's conceptual understanding based on spoken responses. Traditional assessments mainly focus on written examinations, which often fail to measure communication skills and verbal understanding. This project addresses that limitation by automatically analyzing a student's spoken explanation using Artificial Intelligence and Natural Language Processing techniques.

The application accepts an audio file in MP3 or WAV format from the user. The uploaded speech is converted into text using OpenAI's Whisper speech recognition model. The generated transcript is then compared with a predefined reference answer using the Sentence-BERT (SBERT) semantic similarity model. Unlike traditional keyword matching methods, semantic similarity evaluates the actual meaning of the student's response, providing a more accurate assessment.

In addition to semantic analysis, the system extracts important speech characteristics such as speech duration, speech rate, and voice energy using the Librosa library. These audio features help evaluate the quality and fluency of the spoken response. Based on semantic similarity and audio analysis, the system calculates an overall understanding score and provides feedback such as Strong Understanding, Moderate Understanding, or Needs Improvement.

The developed application features a user-friendly interface built with Streamlit. After the analysis, users can download a detailed PDF report containing the transcript, similarity score, audio features, final evaluation, and feedback. The system demonstrates the practical application of Artificial Intelligence in modern education by automating concept evaluation and reducing manual assessment effort.

1. Introduction

Artificial Intelligence has transformed many sectors, including healthcare, finance, transportation, and education. In recent years, educational institutions have increasingly adopted AI-based tools to improve learning experiences and automate assessment processes. Traditional examinations mainly evaluate written answers and often overlook a student's verbal communication skills and conceptual understanding. As a result, there is a growing need for intelligent systems that can evaluate spoken responses accurately and efficiently.

The Voice Based Concept Understanding Analyser is designed to analyze a student's spoken explanation of a given concept. Instead of relying solely on keyword matching, the system uses advanced Natural Language Processing (NLP) techniques to understand the semantic meaning of the student's response. This enables a more reliable evaluation of conceptual understanding.

The application first converts speech into text using the Whisper speech recognition model developed by OpenAI. The generated transcript is then compared with a predefined reference answer using the Sentence-BERT model, which computes semantic similarity between two pieces of text. This approach provides a meaningful comparison based on context rather than exact word matching.

To further improve the evaluation process, the system extracts audio features such as speech duration, speech rate, and energy from the uploaded audio using the Librosa library.

The entire application is implemented using Python and Streamlit, providing an interactive and user-friendly interface. Users can upload audio files, view transcripts, analyze similarity scores, visualize audio waveforms, and download comprehensive PDF reports summarizing the evaluation results.

1.1 Motivation

The primary motivation behind this project is to improve the assessment of students by evaluating both their conceptual understanding and verbal communication skills. Manual evaluation of spoken answers is time-consuming, subjective, and may vary between evaluators. By automating this process using AI, the system ensures faster, more consistent, and objective assessment while providing immediate feedback to learners.

1.2 Applications

The proposed system can be used in various domains, including:

- Online learning platforms
- Educational institutions
- Virtual classrooms
- AI-based interview assessment
- Language learning applications
- Corporate employee training

2. Problem Statement

In the modern education system, evaluating a student's conceptual understanding through spoken responses remains a significant challenge. Traditional assessment methods primarily focus on written examinations or manual interviews, which are often time-consuming, subjective, and inconsistent. Manual evaluation depends heavily on the examiner's judgment and may vary from one evaluator to another, making the assessment process less reliable.

Although several online learning platforms provide quizzes and written assignments, they do not effectively evaluate a student's verbal communication skills or spoken understanding of concepts. Existing speech recognition systems can convert speech into text but generally do not measure how well the spoken content matches the expected concept. As a result, educators find it difficult to assess whether students truly understand a topic based on their verbal explanations.

There is a need for an intelligent system that can automatically convert speech into text, analyze the semantic meaning of the spoken response, evaluate communication quality, and generate a comprehensive performance report. Such a system can improve the efficiency, consistency, and accuracy of student assessments while reducing the workload of educators.

The Voice Based Concept Understanding Analyser Using Artificial Intelligence addresses this challenge by integrating Speech Recognition, Natural Language Processing (NLP), and Machine Learning techniques to provide automated and objective evaluation of students' spoken responses.

3. Need for the Project

The increasing adoption of online education and digital learning platforms has created a demand for intelligent assessment systems capable of evaluating students beyond traditional written examinations. Teachers often face difficulties in assessing spoken responses manually, particularly in large classrooms and virtual learning environments.

The proposed project addresses this challenge by introducing an automated AI-based evaluation system that measures both conceptual understanding and communication skills. It reduces manual effort, improves consistency in scoring, and provides instant feedback to learners.

This project also demonstrates the practical application of Artificial Intelligence, Natural Language Processing, Speech Recognition, and Machine Learning in education. It supports modern learning environments by enabling fair, efficient, and scalable assessment of spoken responses.

4. Objectives of the Project

4.1 Primary Objective

The primary objective of the Voice Based Concept Understanding Analyser Using Artificial Intelligence is to develop an intelligent system capable of automatically evaluating a student's conceptual understanding based on their spoken responses. The system aims to utilize Artificial Intelligence, Speech Recognition, and Natural Language Processing techniques to provide accurate, objective, and efficient assessment while minimizing manual intervention.

4.2 Specific Objectives

- To develop a web-based application using Streamlit for uploading and analyzing audio files.
- To convert speech into text accurately using OpenAI Whisper.
- To compare the student's spoken response with a predefined reference answer using Sentence-BERT semantic similarity.
- To extract important speech characteristics such as duration, speech rate, and energy using Librosa.
- To calculate an overall understanding score based on semantic similarity and audio features.
- To provide meaningful feedback such as Strong Understanding, Moderate Understanding, and Needs Improvement.
- To generate a detailed PDF report containing the complete evaluation results.
- To provide a simple and user-friendly interface for educators and students.
- To reduce manual effort and improve the consistency of spoken response evaluation.

5. Scope of the Project

5.1 Scope

The scope of the proposed system is focused on evaluating conceptual understanding through spoken explanations. The application can be used in educational institutions, online learning platforms, training organizations, and interview assessment systems.

The project mainly covers:

- Speech-to-text conversion.
- Semantic comparison of spoken answers.
- Audio feature extraction.
- Automated scoring.
- Feedback generation.
- PDF report generation.
- Audio waveform visualization.
- Web-based interface using Streamlit.

The current system supports English audio files in MP3 and WAV formats. It can easily be extended to support additional languages and more advanced evaluation techniques in future versions.

5.2 Applications

- Educational Institutions
- Online Learning Platforms
- Virtual Classrooms
- Viva Examination Systems
- Employee Training Programs
- AI-based Interview Assessment
- Language Learning Applications
- Certification Programs
- Corporate Skill Evaluation
- Remote Learning Environments

6. Feasibility Study

A feasibility study determines whether the proposed project is practical and achievable. The project has been analyzed from technical, economic, operational, and schedule perspectives.

6.1 Technical Feasibility

The project is technically feasible because all the required technologies and software tools are open source and readily available.

Technologies used include: Python, Streamlit, OpenAI Whisper, Sentence Transformers, Librosa, ReportLab, and Matplotlib.

These libraries provide the necessary functionalities for speech recognition, semantic analysis, audio processing, visualization, and report generation. The project can be developed and executed using a standard personal computer without requiring specialized hardware.

6.2 Economic Feasibility

The project is economically feasible because it uses free and open-source software tools. No commercial software licenses are required. The development cost is minimal since the project utilizes freely available Python libraries and open-source AI models.

6.3 Operational Feasibility

The application provides a simple graphical user interface developed using Streamlit. Users only need to upload an audio file and click the Analyze Audio button. The generated transcript, evaluation score, waveform, and PDF report are displayed automatically. Therefore, the application is easy to use even for users with limited technical knowledge.

6.4 Schedule Feasibility

The project can be completed within the planned internship duration. The development process follows a modular approach including Requirement Analysis, Design, Development, Testing, Deployment, and Documentation. This ensures timely completion of the project.

7. Project Planning

The project was developed in multiple phases to ensure systematic implementation.

Phase 1 – Requirement Analysis

- Understanding project objectives.
- Identifying software requirements.
- Selecting AI models.

Phase 2 – System Design

- Designing application architecture.
- Planning module interactions.
- Designing Streamlit interface.

Phase 3 – Development

Implementation of individual modules:

- Speech-to-Text Module
- Semantic Analysis Module
- Audio Feature Extraction Module
- Scoring Module
- PDF Report Module
- Streamlit Interface

Phase 4 – Testing

- Module Testing
- Integration Testing
- Output Verification
- Bug Fixing

Phase 5 – Deployment

- GitHub Repository
- Documentation
- Final Project Submission

8. Project Development Methodology

The project follows the Agile Software Development Methodology. The project development was divided into small modules that were independently implemented and tested before integration.

The Agile methodology offers several advantages:

- Continuous testing
- Easy debugging
- Modular development
- Incremental improvements
- Faster delivery
- Better project management

Each module was developed separately and later integrated into the final application.

9. Software Requirements Specification (SRS)

9.1 Purpose

The purpose of this project is to develop an AI-based web application that evaluates a student's conceptual understanding through spoken responses. The application automatically converts speech into text, compares the spoken response with a reference answer, extracts audio features, calculates an understanding score, and generates a detailed PDF report.

9.2 Scope of the Software

The software allows users to upload an audio file and automatically performs the following tasks:

- Speech-to-text conversion.
- Semantic similarity analysis.
- Audio feature extraction.
- Final score calculation.
- Feedback generation.
- Waveform visualization.
- PDF report generation.

The software is intended for educational institutions, online learning platforms, trainers, and interview evaluation systems.

9.3 Functional Requirements

The functional requirements describe what the system should do.

FR-1 User Interface

- The system shall provide a user-friendly web interface.
- Users shall be able to upload MP3 and WAV audio files.

FR-2 Audio Upload

- The system shall allow users to upload audio files.
- Uploaded files shall be stored temporarily in the audio folder.

FR-3 Speech-to-Text

- The system shall convert uploaded speech into text using OpenAI Whisper.

FR-4 Semantic Analysis

- The system shall compare the generated transcript with the reference answer.
- Sentence-BERT shall calculate semantic similarity.

FR-5 Audio Feature Extraction

The system shall calculate:

- Speech Duration
- Speech Rate
- Average Energy

FR-6 Final Evaluation

The system shall calculate:

- Semantic Score
- Final Understanding Score
- Feedback Category

FR-7 Waveform Generation

The application shall display the waveform of the uploaded audio.

FR-8 PDF Report

The application shall generate a downloadable PDF report containing:

- Transcript
- Similarity Score
- Audio Features
- Final Score

- Feedback

9.4 Non-Functional Requirements

These requirements define the quality characteristics of the system.

Performance

- The system should analyze an audio file within a few seconds depending on file size.
- The application should efficiently process audio up to several minutes long.

Reliability

- The application should produce consistent results for the same audio input.
- The system should minimize errors during speech recognition and evaluation.

Usability

- The interface should be simple and intuitive.
- Users should be able to use the application without technical knowledge.

Maintainability

- The project follows a modular architecture.
- Individual modules can be updated independently.

Portability

The application can run on: Windows, Linux, macOS.

Scalability

Future enhancements may include:

- Multiple language support
- Real-time speech analysis
- Cloud deployment
- Multi-user access

10. Hardware Requirements

The hardware requirements for the successful execution of the project are listed below.

Component	Specification
Processor	Intel Core i5 or above
RAM	Minimum 8 GB
Storage	Minimum 20 GB Free Space
Operating System	Windows 10/11, Linux, macOS
Internet	Required for first-time model download
Audio Device	Microphone (optional for future enhancement)

11. Software Requirements

The following software components are required.

Software	Version
Python	3.12+
Streamlit	Latest
OpenAI Whisper	Latest
Sentence Transformers	Latest
Librosa	Latest
NumPy	Latest
Matplotlib	Latest
ReportLab	Latest
Torch	Latest
FFmpeg	Installed and configured
Visual Studio Code	Latest
Git	Latest
GitHub	Repository Hosting

12. Python Libraries Used

The following Python libraries were used during development.

Library	Purpose
Streamlit	User Interface
OpenAI Whisper	Speech Recognition
Sentence Transformers	Semantic Similarity
Torch	AI Model Execution
Librosa	Audio Processing
NumPy	Numerical Computation
Matplotlib	Waveform Visualization
ReportLab	PDF Report Generation
SoundFile	Audio Reading
Scikit-learn	Machine Learning Utilities
Transformers	NLP Support

13. Technologies Used

The Voice Based Concept Understanding Analyser Using Artificial Intelligence is developed using modern Artificial Intelligence, Machine Learning, Natural Language Processing, and Web Application technologies. Each technology plays an important role in the implementation of the project.

13.1 Python

Introduction

Python is a high-level, interpreted, and object-oriented programming language widely used for Artificial Intelligence, Machine Learning, Data Science, and Web Development. It provides a simple syntax and a rich collection of libraries that simplify software development. Python acts as the core programming language for this project.

Features

- Easy to learn and implement
- Large AI and ML ecosystem
- Open-source
- Cross-platform support
- Extensive library support

Role in the Project

- Develop all project modules
- Process uploaded audio files
- Perform semantic analysis
- Calculate scores
- Generate PDF reports
- Build the Streamlit application

13.2 Streamlit

Introduction

Streamlit is an open-source Python framework used for developing interactive web applications with minimal code. It enables developers to quickly create user interfaces for AI and Machine Learning applications.

Features

- Interactive web interface
- Easy deployment

- Fast development
- Supports charts and multimedia
- Python-based framework

Role in the Project

- Upload audio files
- Display transcript
- Show similarity score
- Display waveform
- Show evaluation score
- Download PDF report

13.3 OpenAI Whisper

Introduction

OpenAI Whisper is an advanced Automatic Speech Recognition (ASR) model developed by OpenAI. It converts spoken language into written text with high accuracy. Whisper supports multiple languages and performs well even with noisy audio.

Features

- High transcription accuracy
- Noise tolerant
- Multilingual support
- Open-source AI model

Role in the Project

- Reading uploaded audio
- Converting speech into text
- Producing transcript used for semantic comparison

Without Whisper, the application cannot analyze spoken responses.

13.4 Sentence-BERT (SBERT)

Introduction

Sentence-BERT (SBERT) is a Natural Language Processing model that converts sentences into numerical vector representations called embeddings. Instead of matching words, SBERT compares the meaning of two sentences.

Features

- Semantic similarity

- Sentence embeddings
- Context understanding
- High accuracy

Role in the Project

- Encodes the reference answer and the student transcript, then computes the similarity percentage used in the final evaluation.

Example: Reference — "Artificial Intelligence helps students learn effectively." Student — "AI improves education by making learning easier." Although the wording is different, SBERT recognizes that both sentences have similar meanings.

13.5 Librosa

Introduction

Librosa is a Python library used for audio and music signal processing. It provides tools for extracting audio features.

Features

- Audio loading
- Speech duration
- Waveform generation
- RMS Energy
- Signal processing

Role in the Project

- Duration
- Speech Rate
- Average Energy
- Waveform

These features are included in both the Streamlit application and the generated PDF report.

13.6 NumPy

Introduction

NumPy is a numerical computing library for Python. It provides efficient mathematical operations on arrays and matrices.

Features

- Mathematical operations
- Fast computations

- Array processing
- Statistical functions

Role in the Project

- Calculate average audio energy
- Process numerical values
- Support Librosa computations

13.7 Matplotlib

Introduction

Matplotlib is a Python visualization library used to generate graphs and plots.

Features

- Line plots
- Charts
- Signal visualization
- Image generation

Role in the Project

- Matplotlib is used to generate the waveform image of the uploaded audio. The generated waveform is displayed in the Streamlit application and can also be included in reports.

13.8 ReportLab

Introduction

ReportLab is a Python library used for creating PDF documents programmatically.

Features

- PDF generation
- Text formatting
- Tables
- Images
- Professional reports

Role in the Project

- Transcript
- Semantic Similarity
- Audio Features
- Final Score

- Feedback

This allows users to download and save the evaluation results.

13.9 FFmpeg

Introduction

FFmpeg is an open-source multimedia framework used to process audio and video files.

Features

- Audio conversion
- Format support
- Audio decoding
- Fast processing

Role in the Project

- Whisper uses FFmpeg internally to read uploaded audio files before transcription. It supports MP3 and WAV formats and converts them into the format required by the speech recognition model.

13.10 Git

Introduction

Git is a distributed version control system used for tracking changes in source code.

Features

- Version control
- Branching
- Collaboration
- Backup

Role in the Project

- Track source code changes
- Manage project versions
- Collaborate with team members

13.11 GitHub

Introduction

GitHub is a cloud-based platform for hosting Git repositories.

Features

- Repository hosting
- Collaboration
- Issue tracking
- Documentation
- Continuous Integration support

Role in the Project

- Store project source code
- Share the project with team members
- Maintain project documentation
- Enable collaboration during development

14. Summary of Technologies

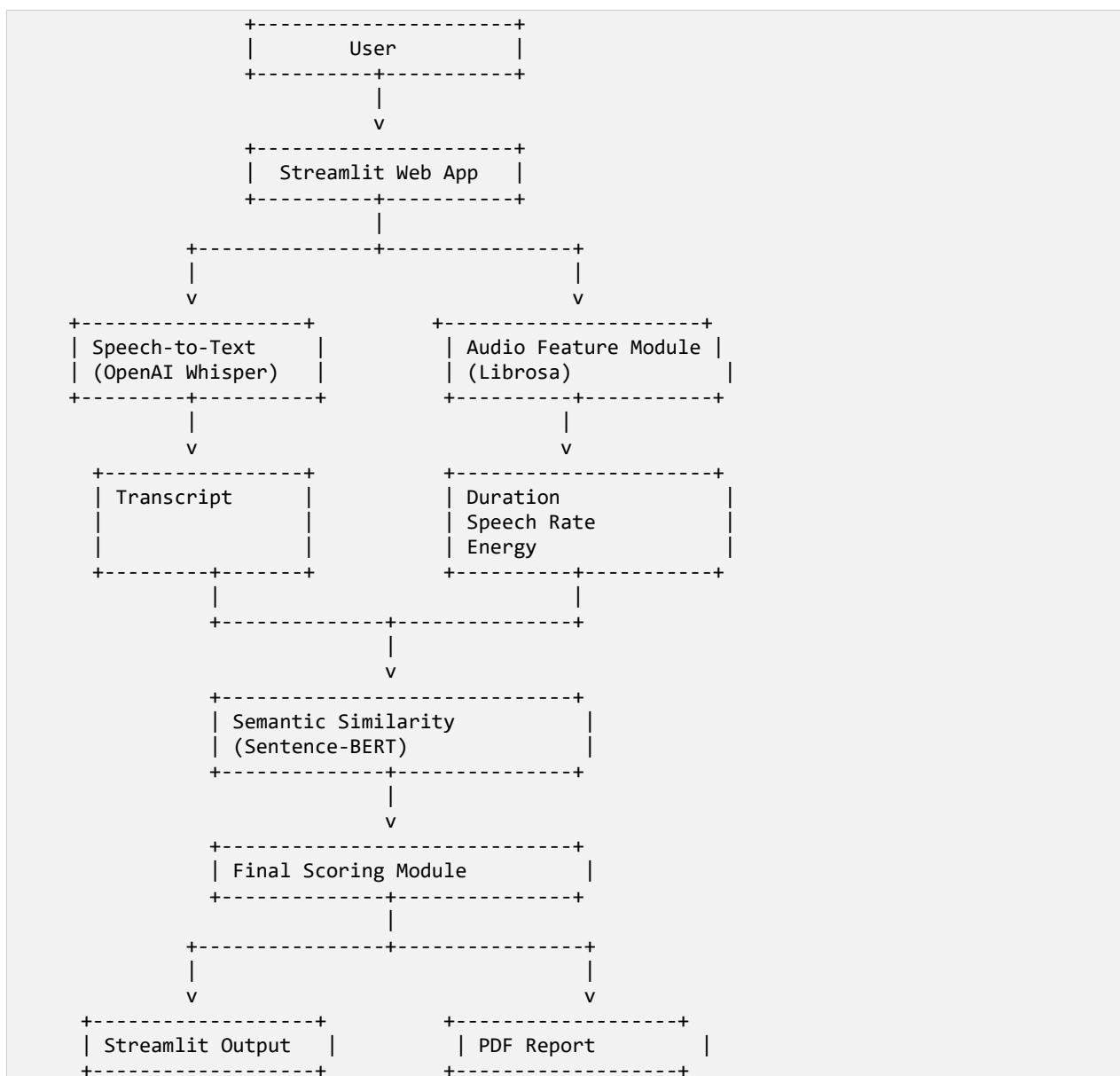
Technology	Purpose
Python	Core programming language
Streamlit	Web application interface
OpenAI Whisper	Speech-to-text conversion
Sentence-BERT	Semantic similarity analysis
Librosa	Audio feature extraction
NumPy	Numerical computations
Matplotlib	Waveform visualization
ReportLab	PDF report generation
FFmpeg	Audio decoding and processing
Git	Version control
GitHub	Repository hosting and collaboration

15. System Architecture

15.1 System Architecture

The Voice Based Concept Understanding Analyser Using Artificial Intelligence follows a modular architecture where each module performs a specific task in the evaluation process. The application begins by accepting an audio file from the user through the Streamlit interface. The generated transcript is compared with the predefined reference answer using Sentence-BERT to calculate semantic similarity. Simultaneously, Librosa extracts audio features such as duration, speech rate, and energy. Finally, a PDF report is created using ReportLab, allowing users to download the complete evaluation report.

System Architecture Diagram



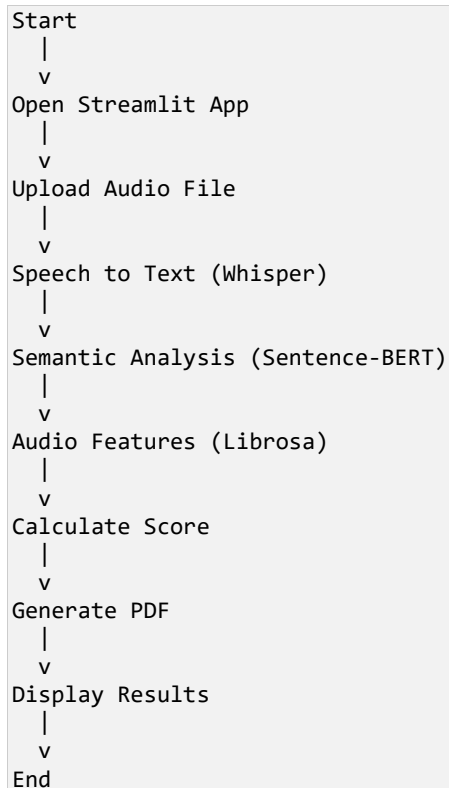
16. Workflow of the System

16.1 Workflow

The workflow of the application consists of the following steps:

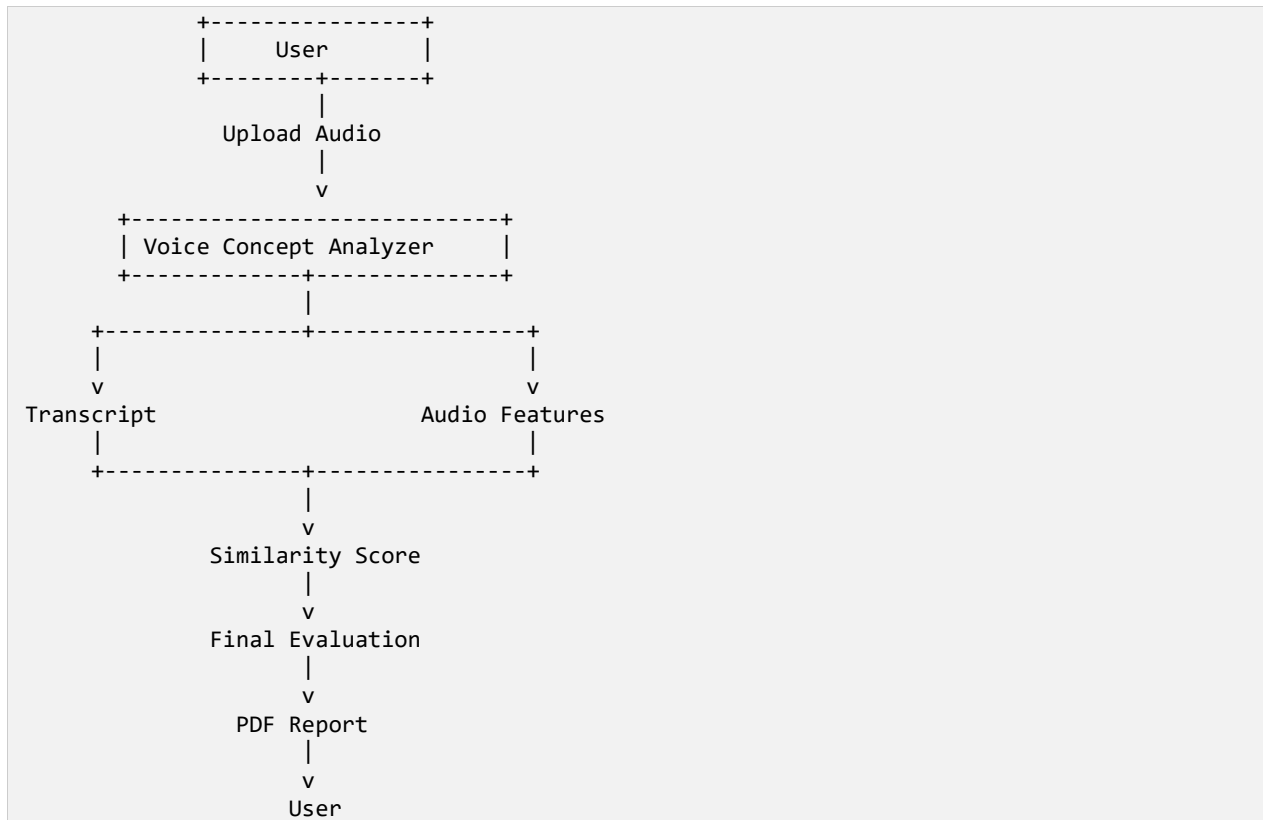
- 1. The user opens the Streamlit application.
- 2. The user uploads an MP3 or WAV audio file.
- 3. The audio file is temporarily stored in the audio folder.
- 4. Whisper converts speech into text.
- 5. Sentence-BERT compares the transcript with the reference answer.
- 6. Librosa extracts duration, speech rate, and energy.
- 7. The scoring module calculates the overall understanding score.
- 8. The application displays the transcript, similarity score, waveform, and feedback.
- 9. A PDF report is generated and made available for download.

Workflow Diagram



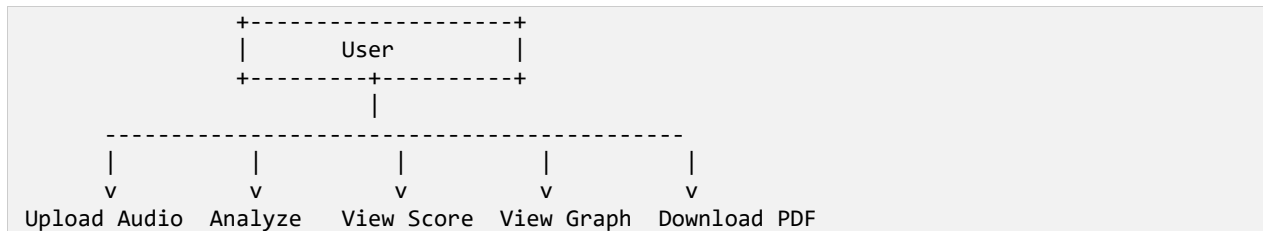
17. Data Flow Diagram (DFD)

17.1 Level-0 DFD



18. Use Case Diagram

Use Case Diagram



19. Module Description

The Voice Based Concept Understanding Analyser Using Artificial Intelligence is divided into several independent modules. Each module performs a specific function, making the system modular, maintainable, and easy to enhance.

The major modules of the project are:

- Streamlit User Interface Module
- Speech-to-Text Module
- Semantic Analysis Module
- Audio Feature Extraction Module
- Scoring Module
- PDF Report Generation Module

19.1 Streamlit User Interface Module

Description

The Streamlit User Interface serves as the front-end of the application. It allows users to interact with the system through a simple web interface. The user uploads an audio file, initiates the analysis process, views the evaluation results, and downloads the generated PDF report.

Responsibilities

- Display project title
- Upload MP3/WAV files
- Play uploaded audio
- Display transcript
- Display semantic similarity
- Display audio features
- Display waveform
- Display final score
- Download PDF report

Input / Output

Input: MP3 Audio, WAV Audio. Output: Transcript, Similarity Score, Audio Features, Waveform, Final Score, PDF Report.

Advantages

- Easy to use
- Interactive interface

- Real-time output
- Minimal user effort

19.2 Speech-to-Text Module

Description

This module converts spoken audio into text using the OpenAI Whisper speech recognition model. After the user uploads an audio file, Whisper processes the audio and produces a transcript that accurately represents the spoken content. The generated transcript becomes the input for the semantic analysis module.

Process

- Load Whisper Model
- Read Audio
- Convert Speech into Text
- Generate Transcript

Example output: "Artificial intelligence is transforming education by helping students learn more effectively."

Advantages

- High transcription accuracy
- Supports multiple audio formats
- Noise-resistant speech recognition

19.3 Semantic Analysis Module

Description

The Semantic Analysis Module compares the generated transcript with the predefined reference answer. Unlike keyword matching systems, this module understands the meaning of sentences using Sentence-BERT (SBERT). It generates sentence embeddings and computes cosine similarity to determine how closely the student's explanation matches the expected answer.

Process

Reference Answer → Sentence Embedding → Student Transcript → Sentence Embedding → Cosine Similarity → Similarity Percentage

Example: Similarity = 84.01%

Advantages

- Context-aware comparison
- Better than keyword matching

- High semantic accuracy

19.4 Audio Feature Extraction Module

Description

This module analyzes the uploaded audio signal using the Librosa library. It extracts several important speech characteristics that help evaluate communication quality. The extracted features are later used during final scoring.

Features Extracted

- Duration – measures the total length of the audio. Example: 16.73 Seconds
- Speech Rate – calculated as $\text{Words Spoken} \div \text{Duration} \times 60$. Example: 121.95 Words Per Minute
- Energy – measures average voice intensity using RMS Energy. Example: 0.0666

Advantages

- Fast processing
- Accurate calculations
- Supports multiple audio formats

19.5 Audio Waveform Module

Description

The waveform module generates a graphical representation of the uploaded audio. It helps visualize speech intensity and variations throughout the recording. The waveform is displayed within the Streamlit application.

Process

Audio → Read Signal → Generate Waveform → Save Image → Display in Streamlit

Advantages

- Better visualization
- Easy interpretation
- Professional interface

19.6 Scoring Module

Description

The Scoring Module combines semantic similarity and audio features to calculate the overall understanding score. The scoring system considers Semantic Similarity, Speech Rate, and Voice Energy. Each parameter contributes to the final evaluation.

Scoring Formula

Component	Weight
Semantic Similarity	80%
Speech Rate	10%
Energy	10%

Feedback Categories

Final Score	Feedback
85 – 100	Strong Understanding
65 – 84	Moderate Understanding
Below 65	Needs Improvement

19.7 PDF Report Generation Module

Description

The PDF Report Module automatically generates a downloadable report after the evaluation process. The report provides a permanent record of the student's performance.

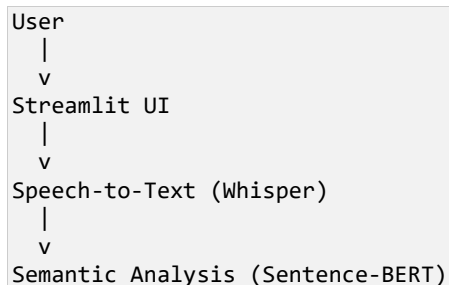
PDF Contents

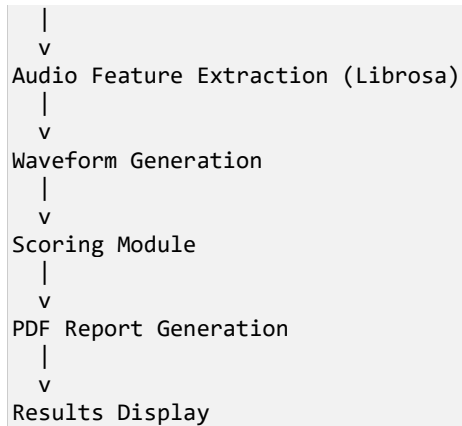
- Project Title
- Transcript
- Semantic Similarity
- Audio Duration
- Speech Rate
- Energy
- Final Score
- Feedback

Advantages

- Downloadable
- Professional formatting
- Easy sharing
- Permanent record

19.8 Module Interaction





20. Algorithms Used

The Voice Based Concept Understanding Analyser Using Artificial Intelligence uses several Artificial Intelligence, Natural Language Processing (NLP), and Digital Signal Processing algorithms to evaluate a student's spoken response. These algorithms work together to provide an accurate and automated assessment.

The major algorithms and models used in this project are:

- Whisper Speech Recognition Algorithm
- Sentence-BERT Semantic Similarity Algorithm
- Cosine Similarity Algorithm
- Audio Feature Extraction Algorithm
- Scoring Algorithm

20.1 Whisper Speech Recognition Algorithm

Introduction

Whisper is an Automatic Speech Recognition (ASR) model developed by OpenAI. It converts spoken language into text using deep learning techniques based on Transformer architecture. The model is trained on a large multilingual dataset, enabling it to accurately transcribe speech even in noisy environments.

Working Principle

- Accept audio input.
- Preprocess the audio using FFmpeg.
- Convert the audio into Mel Spectrograms.
- Process the spectrogram using the Whisper neural network.
- Generate the transcript.

Algorithm

Input: Audio File
Output: Transcript

Step 1: Load Whisper Model
Step 2: Read Audio File
Step 3: Convert Audio into Mel Spectrogram
Step 4: Pass Spectrogram to Whisper Model
Step 5: Decode Speech into Text
Step 6: Return Transcript

Advantages

- High transcription accuracy
- Supports noisy audio
- Multilingual support
- AI-based speech recognition

20.2 Sentence-BERT Algorithm

Introduction

Sentence-BERT (SBERT) is an extension of the BERT language model designed to generate meaningful sentence embeddings. Instead of comparing words, SBERT compares the semantic meaning of complete sentences.

Working Principle

- Convert the reference answer into a sentence embedding.
- Convert the student's transcript into a sentence embedding.
- Compute cosine similarity.
- Return similarity percentage.

Algorithm

Input: Reference Answer, Student Transcript
Output: Similarity Percentage

Step 1: Load Sentence-BERT Model
Step 2: Encode Reference Answer
Step 3: Encode Student Transcript
Step 4: Compute Cosine Similarity
Step 5: Convert Similarity into Percentage
Step 6: Return Similarity Score

Advantages

- Understands sentence meaning
- Context-aware comparison
- Better than keyword matching

20.3 Cosine Similarity Algorithm

Introduction

Cosine Similarity measures the similarity between two vectors by calculating the cosine of the angle between them. In this project, it compares the embeddings generated by Sentence-BERT. The similarity value ranges from 0 to 1.

Formula

Cosine Similarity = $(A \cdot B) / (|A| |B|)$, where A is the Reference Answer Embedding and B is the Student Transcript Embedding.

Advantages

- Fast computation
- High semantic accuracy
- Suitable for NLP tasks

20.4 Audio Feature Extraction Algorithm

Introduction

Audio Feature Extraction is performed using the Librosa library. The algorithm calculates important speech characteristics used for evaluation.

Features

- Duration – measures total audio length.
- Speech Rate – calculated as $(\text{Words} \div \text{Duration}) \times 60$.
- Energy – calculated using RMS (Root Mean Square): $\text{Energy} = \text{Mean}(\text{RMS})$.

Algorithm

```
Input:  Audio File, Transcript
Output: Audio Features

Step 1: Load Audio
Step 2: Calculate Duration
Step 3: Count Number of Words
Step 4: Compute Speech Rate
Step 5: Calculate RMS Energy
Step 6: Return Audio Features
```

Advantages

- Accurate measurements
- Fast processing
- Supports speech evaluation

20.5 Final Scoring Algorithm

Introduction

The Scoring Algorithm combines semantic similarity and communication quality to calculate the final understanding score.

Parameter	Weight
Semantic Similarity	80%
Speech Rate	10%
Energy	10%

Algorithm

Input: Semantic Similarity, Speech Rate, Energy
Output: Final Score, Feedback

Step 1: Calculate Semantic Score
Step 2: Evaluate Speech Rate
Step 3: Evaluate Energy
Step 4: Compute Final Score
Step 5: Assign Feedback
Step 6: Return Final Score and Feedback

Feedback Conditions

If Score ≥ 85 -> Strong Understanding
Else If Score ≥ 65 -> Moderate Understanding
Else -> Needs Improvement

21. Time Complexity

Module	Complexity
Speech-to-Text	$O(n)$
Semantic Similarity	$O(n)$
Audio Feature Extraction	$O(n)$
Score Calculation	$O(1)$
PDF Generation	$O(n)$

22. Summary

The combination of Whisper, Sentence-BERT, Cosine Similarity, Librosa, and the Scoring Algorithm enables the system to evaluate spoken responses intelligently. These algorithms ensure that the application assesses both the content and quality of speech, making the evaluation more accurate and reliable than traditional keyword-based methods.

23. Implementation

23.1 Introduction

The implementation phase involves integrating all the developed modules into a single functional application. The project was implemented using Python and follows a modular programming approach. Each module performs an independent task, making the system easier to understand, test, and maintain.

The application is built using Streamlit, which provides a simple and interactive web interface. Artificial Intelligence models such as OpenAI Whisper and Sentence-BERT are integrated to perform speech recognition and semantic similarity analysis. Additional modules extract audio features, calculate the final evaluation score, generate waveform visualizations, and create PDF reports.

23.2 Project Folder Structure

```
Voice-Based-Concept-Understanding-Analyser/  
|  
+-- app.py  
+-- requirements.txt  
+-- README.md  
+-- .gitignore  
|  
+-- audio/  
|   +-- sample.mp3  
|  
+-- assets/  
|   +-- waveform.png  
|   +-- output1.png  
|   +-- output2.png  
|   +-- output3.png  
|  
+-- reports/  
|   +-- Concept_Report.pdf  
|  
+-- modules/  
|   +-- speech_to_text.py  
|   +-- semantic_analysis.py  
|   +-- audio_features.py  
|   +-- scoring.py  
|   +-- pdf_report.py  
|  
+-- tests/  
|   +-- test_whisper.py  
|   +-- test_semantic.py  
|   +-- test_audio_features.py  
|   +-- test_scoring.py
```

23.3 Module Implementation

1. Streamlit Interface

The Streamlit interface provides an interactive environment where users upload audio files and receive evaluation results.

- Upload audio
- Display transcript
- Show similarity score
- Display audio features
- Show waveform
- Generate PDF
- Download report

2. Speech-to-Text Module

This module uses the Whisper model.

- Load Whisper Base Model
- Accept uploaded audio
- Convert speech into text
- Return transcript

Example output: "Artificial intelligence is transforming education by helping students learn more effectively..."

3. Semantic Analysis Module

The transcript generated by Whisper is compared with the reference answer.

- Load Sentence-BERT
- Encode both sentences
- Calculate cosine similarity
- Return similarity percentage

Example: Similarity Score : 100%

4. Audio Feature Module

The uploaded audio is analyzed using Librosa. Features extracted: Duration, Speech Rate, Energy.

Example output: Duration : 16.73 sec, Speech Rate : 121.95 WPM, Energy : 0.0666

5. Scoring Module

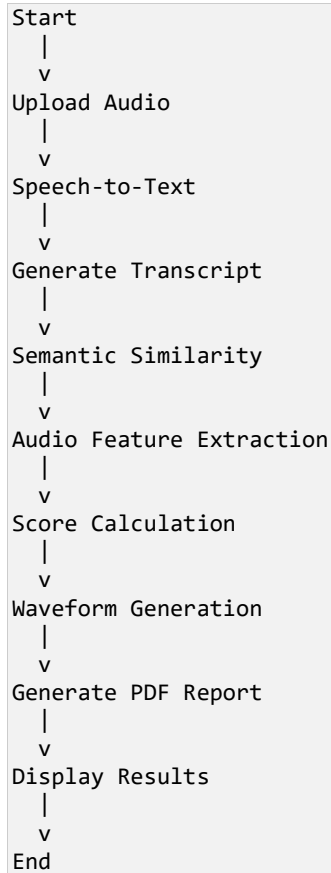
The scoring module combines semantic similarity with communication quality. Weight distribution: Semantic Similarity 80%, Speech Rate 10%, Energy 10%.

Output: Final Score : 100, Feedback : Strong Understanding

6. PDF Report Module

The ReportLab library is used for generating downloadable reports containing the transcript, similarity score, duration, speech rate, energy, final score, and feedback (Concept_Report.pdf).

23.4 Implementation Workflow



23.5 Execution Steps

The following steps are used to execute the project.

Step 1 — Clone the GitHub repository:

```
git clone https://github.com/SwethamaddaC/Voice-Based-Concept-Understanding-Analyser.git
```

Step 2 — Navigate to the project directory:

```
cd Voice-Based-Concept-Understanding-Analyser
```

Step 3 — Create a virtual environment:

```
python -m venv .venv
```

Step 4 — Activate the virtual environment:

```
Windows:      .venv\Scripts\activate
Linux/macOS:  source .venv/bin/activate
```

Step 5 — Install dependencies:

```
pip install -r requirements.txt
```

Step 6 — Run the Streamlit application:

```
streamlit run app.py
```

Step 7 — Open the browser:

```
http://localhost:8501
```

Step 8 — Upload an audio file. Supported formats: MP3, WAV.

Step 9 — Click Analyze Audio. The system automatically converts speech into text, calculates semantic similarity, extracts audio features, calculates the final score, displays the waveform, and generates the PDF report.

23.6 Summary

The implementation phase successfully integrates all functional modules into a single application. The developed system provides a user-friendly interface for uploading audio, analyzing conceptual understanding using AI models, extracting audio features, visualizing the waveform, generating evaluation scores, and producing downloadable PDF reports. The modular implementation makes the project easy to understand, maintain, and extend with additional features in the future.

24. Testing

24.1 Introduction

Testing is a crucial phase in software development that ensures the application functions correctly and meets the specified requirements. In this project, each module was tested individually before integrating them into the complete system. The testing process verified the correctness, reliability, and performance of the application.

The project was tested using sample audio files to ensure that speech recognition, semantic analysis, audio feature extraction, scoring, waveform visualization, and PDF generation worked as expected.

24.2 Unit Testing

Each module was tested independently.

Module	Test Status
Speech-to-Text	Passed
Semantic Analysis	Passed
Audio Features	Passed
Scoring Module	Passed
PDF Report Module	Passed
Waveform Generation	Passed

Test 1 – Speech-to-Text Module

Objective: Verify that Whisper correctly converts speech into text. Input: sample.mp3. Expected Output: Accurate transcript.

Actual Output: "Artificial intelligence is transforming education by helping students learn more effectively. Machine learning enables computers to learn from data without being explicitly programmed. Natural language processing allows computers to understand and generate human language." Result: Passed.

Test 2 – Semantic Similarity Module

Objective: Verify similarity calculation. Input: Reference Answer, Student Transcript. Expected Output: Similarity Percentage. Actual Output: 100.00%. Result: Passed.

Test 3 – Audio Feature Extraction

Objective: Verify extraction of audio characteristics.

Feature	Value
Duration	16.73 sec
Speech Rate	121.95 WPM
Energy	0.0666

Result: Passed

Test 4 – Final Scoring Module

Objective: Verify overall evaluation score.

Parameter	Result
Final Score	100.00
Feedback	Strong Understanding

Result: Passed

Test 5 – PDF Generation

Objective: Verify PDF creation. Output: Concept_Report.pdf generated successfully. Result: Passed.

Test 6 – Waveform Generation

Objective: Verify waveform creation. Output: waveform.png generated successfully. Result: Passed.

24.3 Integration Testing

After testing individual modules, all modules were integrated.

```
Upload Audio
-> Speech Recognition
-> Transcript
-> Semantic Analysis
-> Feature Extraction
-> Score Calculation
-> Waveform
-> PDF Report
-> Display Results
```

Result: All modules communicated successfully. Status: Passed.

24.4 Functional Testing

The application was tested against all functional requirements.

Function	Status
Upload Audio	Passed
Play Audio	Passed
Speech-to-Text	Passed
Semantic Analysis	Passed
Audio Feature Extraction	Passed
Waveform Display	Passed
Final Score	Passed
Feedback Generation	Passed
PDF Download	Passed

24.5 System Testing

The complete application was tested as a whole. Scenario: User uploads an MP3 file. Expected process: upload file, convert speech to text, calculate similarity, extract audio features, generate waveform, calculate score, generate PDF, and display results. Actual Result: All operations completed successfully. Status: Passed.

24.6 User Acceptance Testing (UAT)

The application was evaluated from the user's perspective. User actions: upload audio file, click Analyze Audio, view transcript, view semantic similarity, view audio features, view waveform, and download the PDF report. Observation: The application responded correctly and produced the expected outputs. Status: Accepted.

24.7 Test Cases

Test Case ID	Description	Expected Result	Status
TC-01	Upload MP3 file	File uploaded	Passed
TC-02	Upload WAV file	File uploaded	Passed
TC-03	Convert speech to text	Transcript generated	Passed
TC-04	Calculate semantic similarity	Similarity displayed	Passed
TC-05	Extract audio features	Features displayed	Passed
TC-06	Generate waveform	Waveform displayed	Passed
TC-07	Calculate final score	Score displayed	Passed
TC-08	Generate PDF report	PDF downloaded	Passed

24.8 Sample Output

Transcript: "Artificial intelligence is transforming education by helping students learn more effectively. Machine learning enables computers to learn from data without being explicitly programmed. Natural language processing allows computers to understand and generate human language."

Semantic Similarity: 100.00% | Duration: 16.73 sec | Speech Rate: 121.95 WPM | Energy: 0.0666

Final Score: 100.00 | Feedback: Strong Understanding | PDF Report: Concept_Report.pdf

24.9 Performance Evaluation

Parameter	Observation
Audio Upload	Successful
Whisper Processing	Accurate
Semantic Analysis	Fast
Feature Extraction	Accurate
PDF Generation	Successful
Waveform Creation	Successful
Overall Performance	Good

24.10 Testing Summary

All modules were tested individually and after integration. The application successfully converted speech into text, evaluated semantic similarity, extracted audio features, generated waveform visualizations, calculated the final score, and created a downloadable PDF report. No critical functional errors were observed during testing.

25. Results and Discussion

25.1 Introduction

The developed Voice Based Concept Understanding Analyser Using Artificial Intelligence was successfully implemented and tested. The application accurately converts speech into text, evaluates conceptual understanding using semantic similarity, extracts audio features, calculates an overall score, generates waveform visualizations, and creates downloadable PDF reports.

25.2 Experimental Results

The project was tested using a sample audio file containing an explanation about Artificial Intelligence.

Parameter	Output
Audio Format	MP3
Audio Duration	16.73 sec
Transcript	Generated Successfully
Semantic Similarity	100.00%
Speech Rate	121.95 WPM
Energy	0.0666

Parameter	Output
Final Score	100.00
Feedback	Strong Understanding
PDF Report	Generated Successfully
Waveform	Displayed Successfully

25.3 Output Screens

The following outputs were successfully generated during execution:

Figure 25.3.1: Home Page of the Application

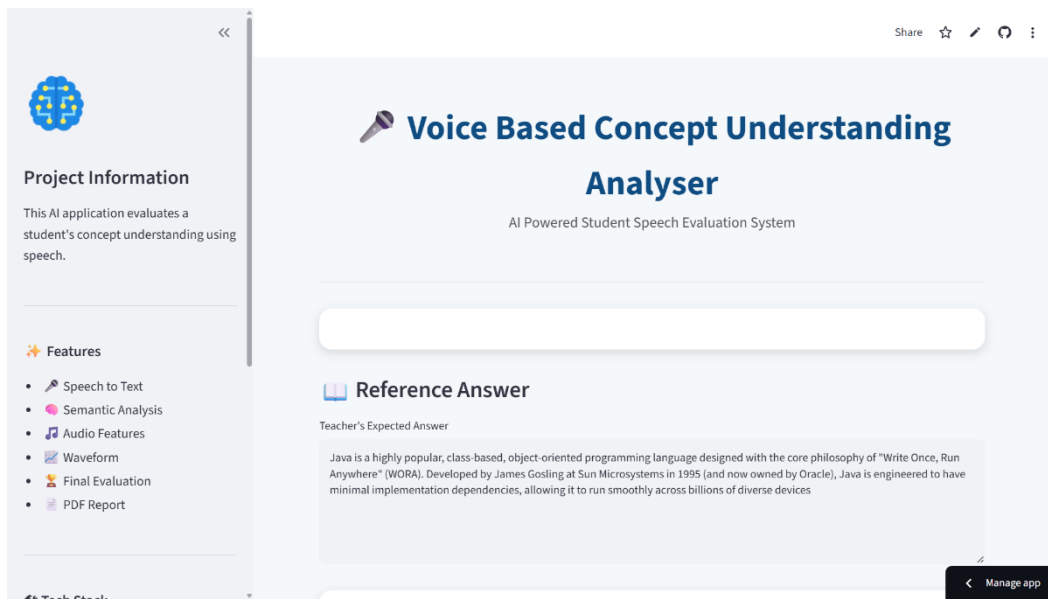


Figure 25.3.2: Audio Upload Screen

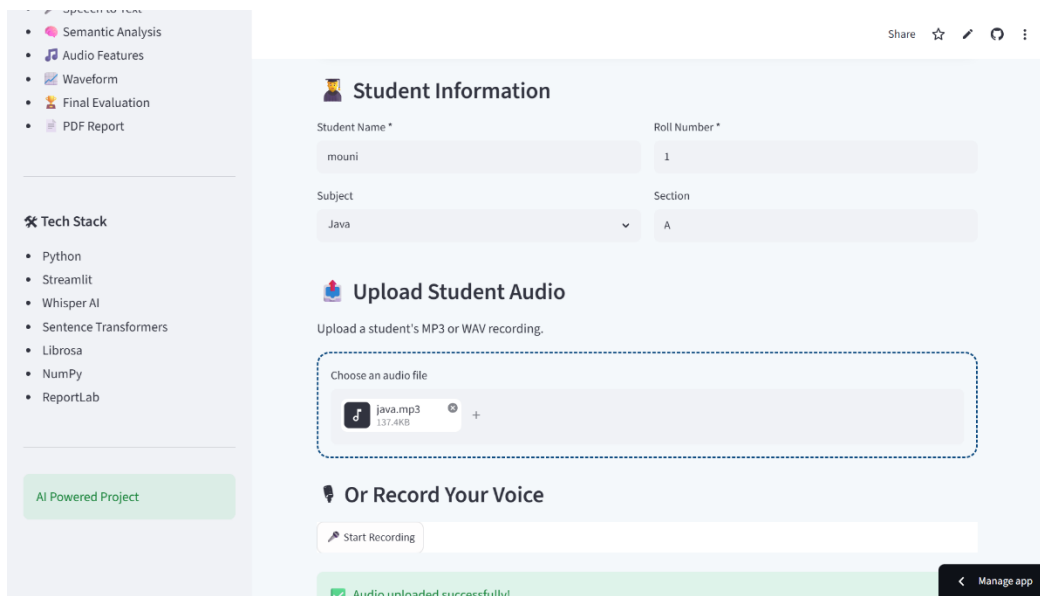


Figure 25.3.3: Transcript Output & Semantic Similarity Result

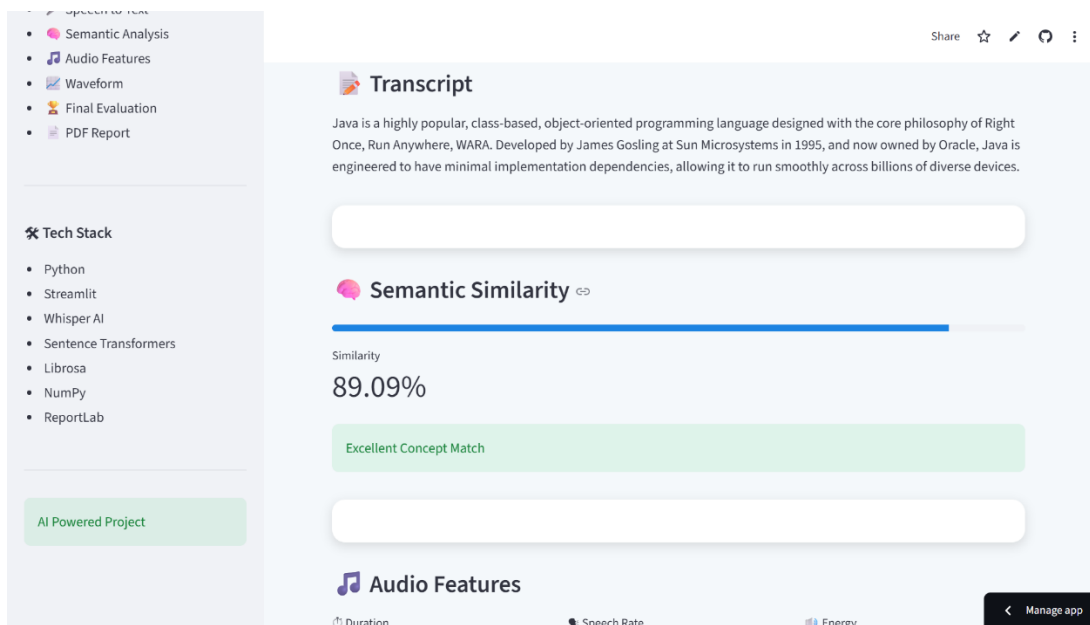


Figure 25.3.4: Audio Features & Waveform Visualization

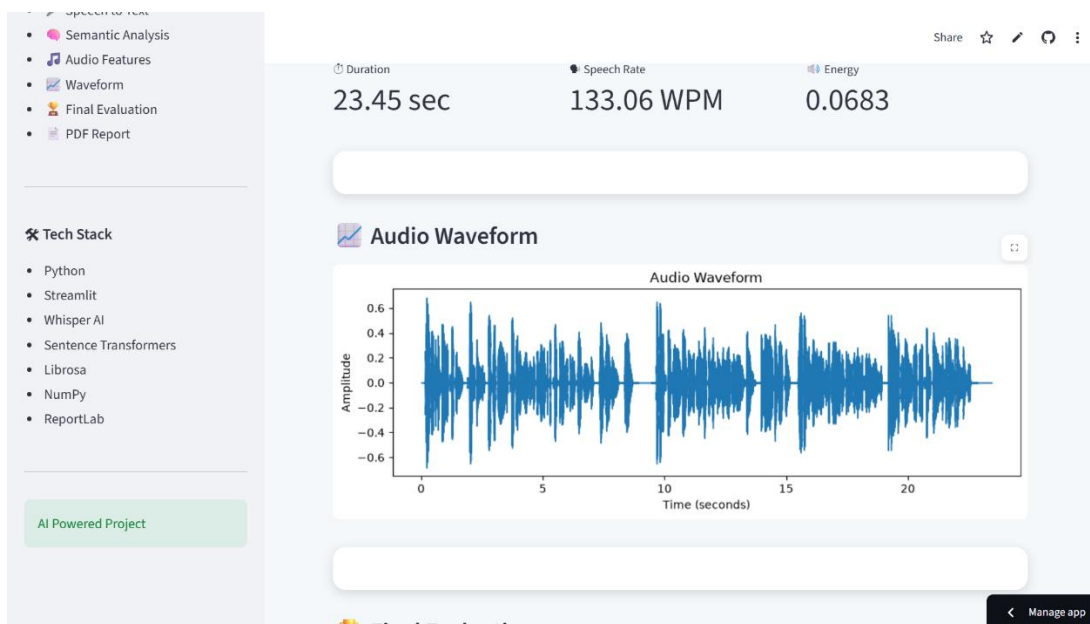


Figure 25.3.5: Final Score

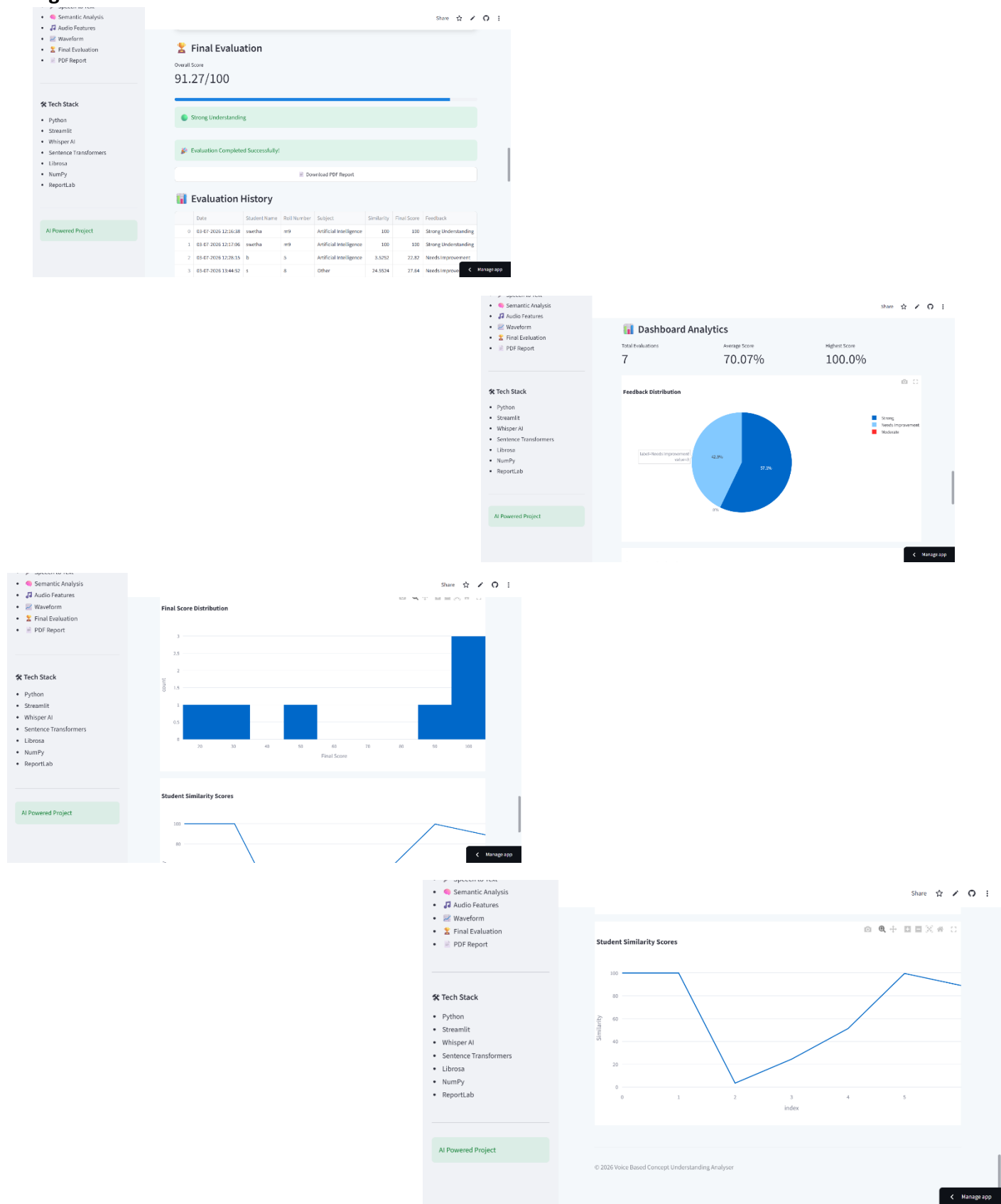
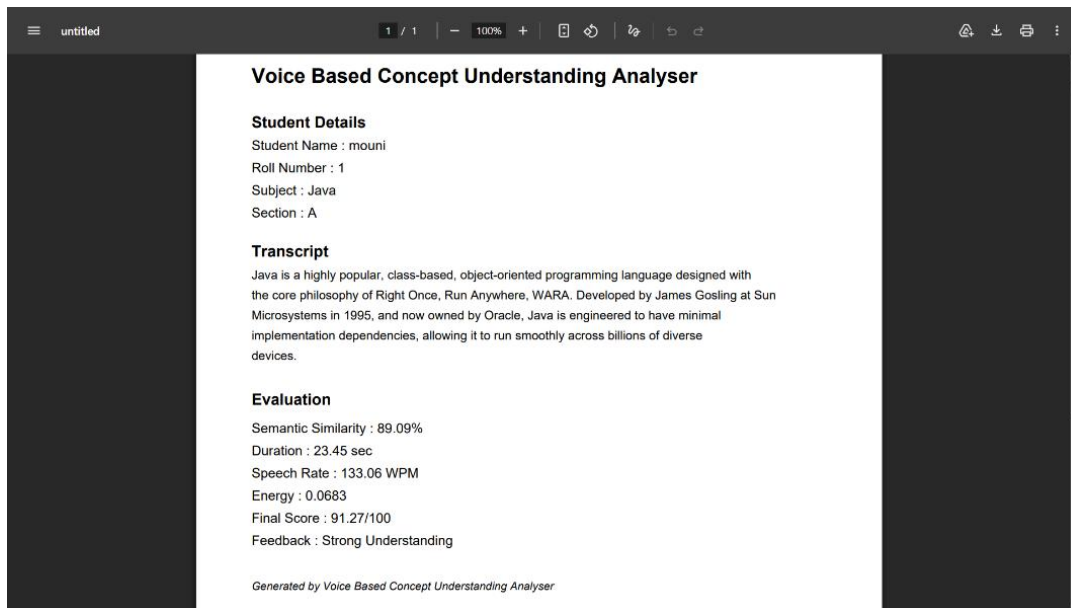


Figure 25.3.6: PDF Report



25.4 Discussion

The proposed system successfully integrates multiple Artificial Intelligence technologies into a single application. The system performs the following tasks efficiently:

- Converts speech into text using Whisper.
- Compares conceptual understanding using Sentence-BERT.
- Extracts speech characteristics using Librosa.
- Calculates an overall evaluation score.
- Displays waveform visualization.
- Generates downloadable PDF reports.

Unlike traditional keyword-based evaluation methods, this application evaluates the meaning of the spoken response, making the assessment more reliable and accurate.

26. Advantages

The proposed system offers several advantages over traditional evaluation methods.

- Automatic speech evaluation.
- AI-based semantic understanding.
- Reduces manual assessment effort.
- Fast and accurate speech recognition.
- Interactive web interface.
- Supports MP3 and WAV audio formats.
- Automatic PDF report generation.
- Waveform visualization for speech analysis.
- Modular and scalable architecture.
- Easy deployment using Streamlit.

27. Limitations

Although the application performs well, it has certain limitations.

- Currently accepts only MP3 and WAV files.
- Uses a predefined reference answer.
- Does not evaluate pronunciation or grammar in detail.
- Performance depends on audio quality.
- Limited to English language evaluation.
- Does not support live microphone recording.
- Whisper processing may take longer on CPU-only systems.

28. Future Enhancements

The project can be enhanced with additional features in future versions.

- Live microphone recording.
- Real-time speech evaluation.
- Multi-language support.
- Pronunciation assessment.
- Grammar and vocabulary evaluation.
- Emotion and sentiment analysis.
- AI-generated personalized feedback.
- Student database integration.
- Cloud deployment using AWS or Azure.
- Mobile application support.
- Dashboard for teachers to monitor student performance.
- Integration with Learning Management Systems (LMS).

28. Applications

This project can be used in various domains.

- Educational institutions.
- Online learning platforms.
- Virtual classrooms.
- Interview assessment systems.
- Corporate training programs.
- Public speaking evaluation.
- Language learning applications.
- AI-based examination systems.
- Employee skill assessment.
- E-learning platforms.

29. Conclusion

The Voice Based Concept Understanding Analyser Using Artificial Intelligence was successfully designed, implemented, and tested. The application demonstrates how Artificial Intelligence can automate the evaluation of spoken responses by combining speech recognition, semantic analysis, audio feature extraction, and scoring techniques.

The integration of OpenAI Whisper for speech-to-text conversion, Sentence-BERT for semantic similarity analysis, Librosa for audio feature extraction, and ReportLab for PDF generation provides a complete solution for concept understanding analysis.

The developed system produces accurate transcripts, meaningful similarity scores, detailed audio feature analysis, waveform visualizations, and downloadable evaluation reports through a user-friendly Streamlit interface.

Overall, the project provides an efficient, intelligent, and scalable approach for assessing spoken conceptual understanding, making it suitable for educational and training environments.

30. References

The following references were used during the development of this project.

- OpenAI. Whisper: Robust Speech Recognition via Large-Scale Weak Supervision. <https://openai.com/research/whisper>
- Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT Networks. EMNLP.
- Librosa Documentation. <https://librosa.org>
- Streamlit Documentation. <https://streamlit.io>
- ReportLab User Guide. <https://www.reportlab.com>
- Hugging Face Sentence Transformers Documentation. <https://www.sbert.net>
- Python Documentation. <https://docs.python.org/3>
- NumPy Documentation. <https://numpy.org>
- Matplotlib Documentation. <https://matplotlib.org>
- GitHub Documentation. <https://docs.github.com>

31. Appendix

Software Requirements

- Windows 10/11
- Python 3.12+
- Streamlit
- OpenAI Whisper
- Sentence-Transformers
- Librosa
- NumPy
- Matplotlib
- ReportLab
- FFmpeg
- Git

Hardware Requirements

- Intel Core i3 or above
- 4 GB RAM (8 GB recommended)
- 2 GB free storage
- Internet connection (for initial model download)

Project Repository

GitHub Repository: <https://github.com/SwethamaddaC/Voice-Based-Concept-Understanding-Analyser>

Final Outcome

The project successfully achieves its objective of automatically evaluating a student's conceptual understanding from spoken responses. By leveraging AI and NLP technologies, it provides a practical solution for educational assessment and demonstrates the real-world application of speech recognition and semantic analysis.